# COMP 424 - Artificial Intelligence
# Fall 2026

Lecture 10: October 7th, 2026

Propositional Logic

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# Announcements

- Midterm help sessions coming soon – check Ed for details!
- Midterm crib sheet, review doc

# The Goal Today



# The Goal Today

- Can you play battleship with
  - Constructive or Local Searches?
  - MiniMax Search?
  - MCTS?

  - AND-OR Search?

We'd like a more expressive kind of knowledge:

- How to **represent** it
- How to **reason** about it → infer what to do next

# A knowledge-based agent

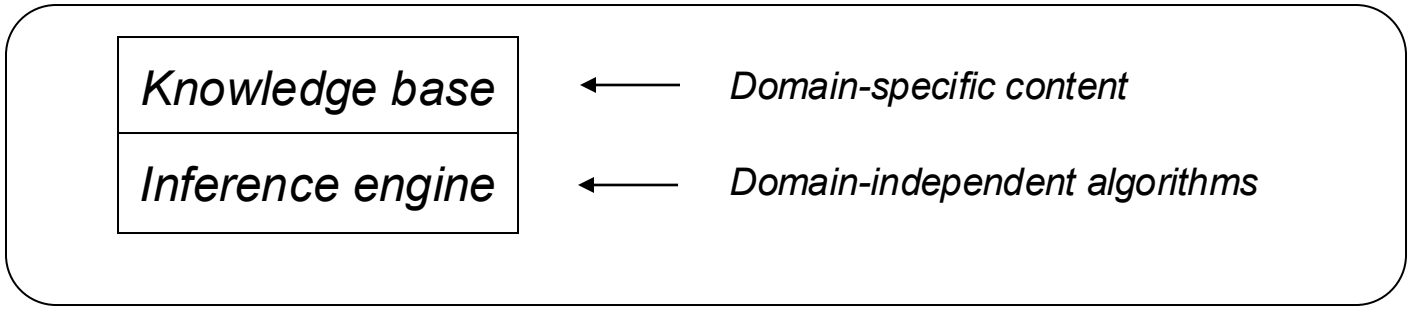
Two key components:

- The **knowledge base**, which contains a set of facts expressed in some formal, standard language
- The **inference engine**, which uses general rules to deduce new facts from those in the KB

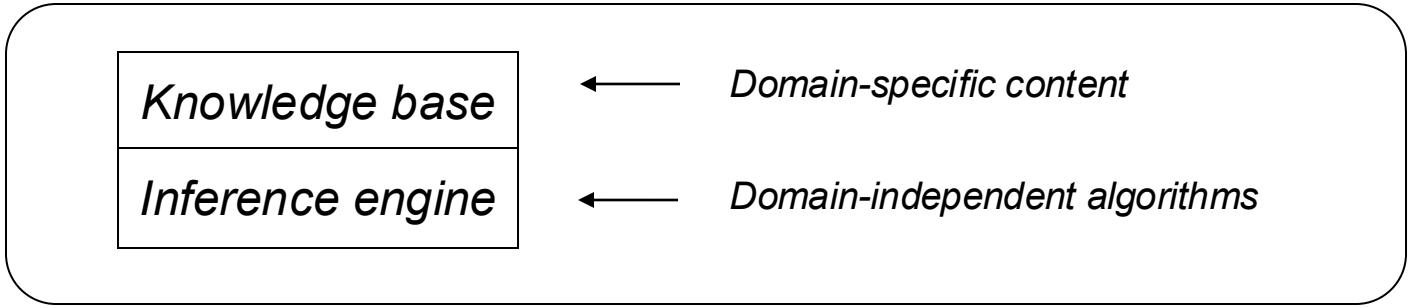
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# A knowledge-based agent




- KB may be initialized with **background knowledge**
- When the agent perceives an observation, it adds the related fact to its KB
- It queries the KB for what action to take in a given state
- It adds the fact that it took the action to the KB
- All the while, the inference engine may be expanding the set of facts in the KB

# An example: Wumpus World

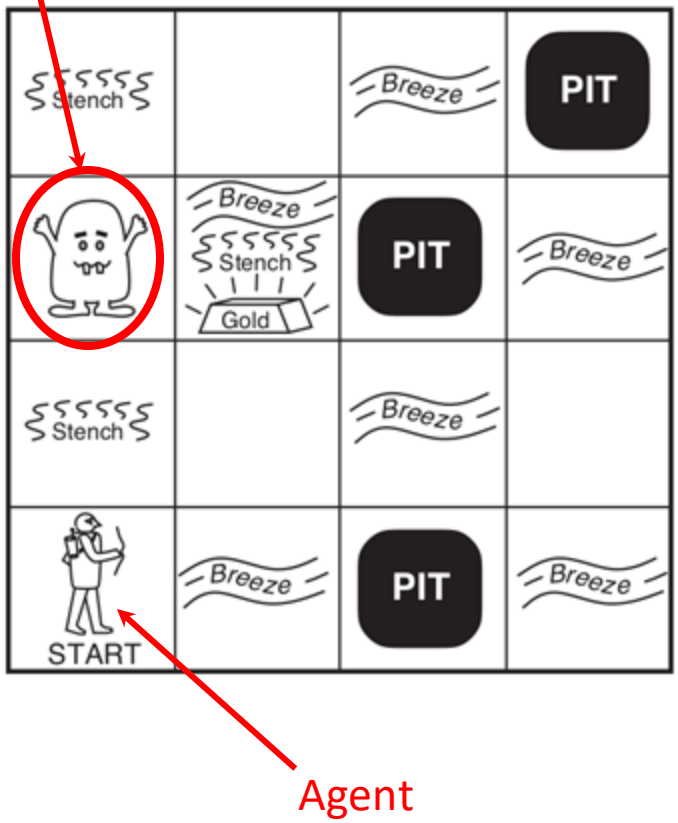
**Percepts**: *Breeze, Glitter, Stench*
(local perception only)

**Actions**: *Turn-left, Turn-right, Forward, Grab, Shoot*

**Goal**: *Get the Gold and return to Start;*
*Don't enter a Pit or alive Wumpus square*

**Rules of the environment:**

- Squares adjacent to Wumpus are smelly (*Stench*)
- Squares adjacent to Pit are breezy (*Breeze*)
- *Glitter* if and only if Gold is in the same square
- *Shooting* kills the Wumpus if you are facing it
- *Shooting* uses up the only Arrow
- *Grabbing* picks up the Gold if in the same square

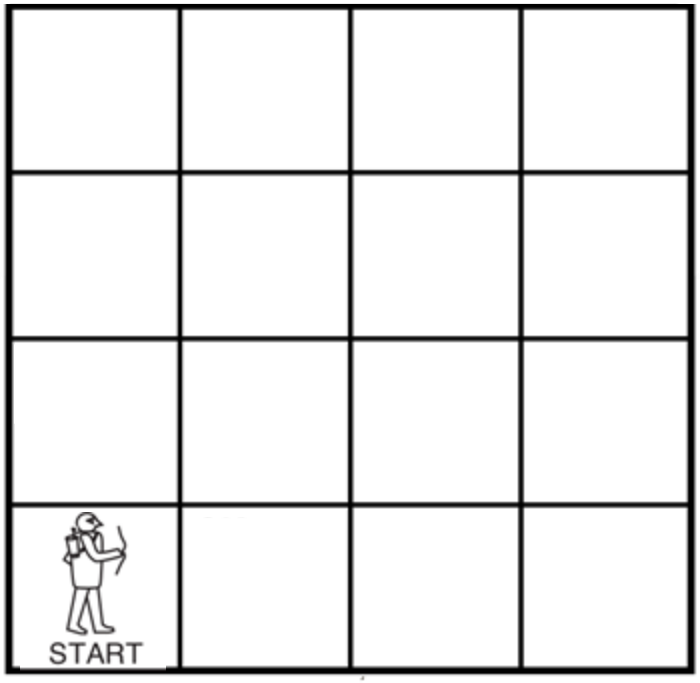


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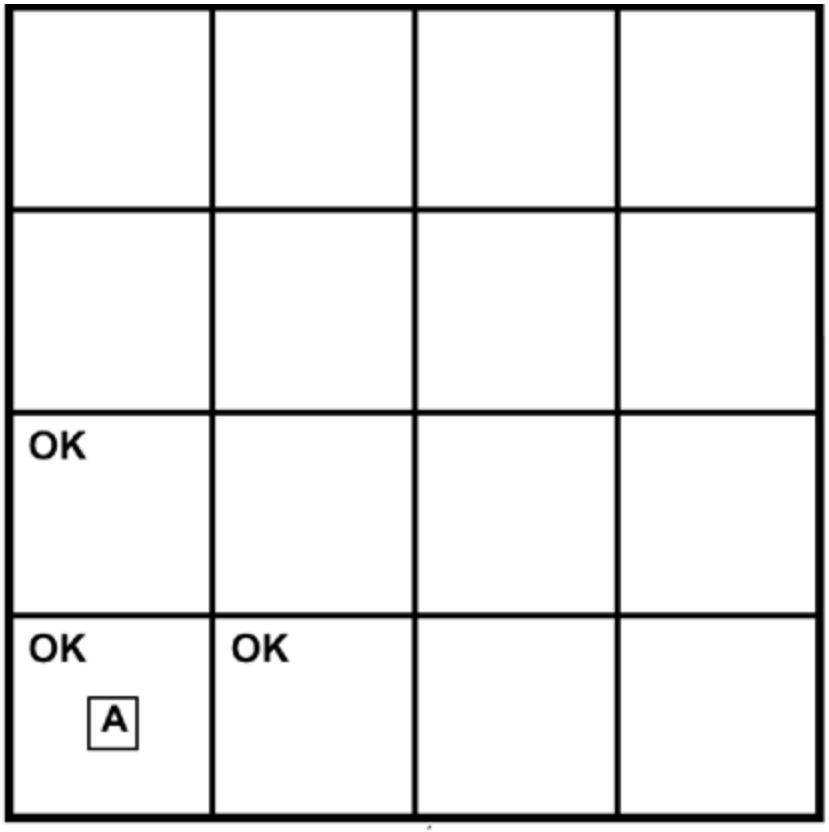
# Wumpus World Characteristics

- The world is **static**: positions of the pits, gold, and monster do not change during the game
- The actions have deterministic effects
- The state is **partially observable**
  - The agent doesn't know the map beforehand and can only perceive what's in the square it occupies
  - → It must **infer** the map from local perception
- What's the belief state for this domain?

# Wumpus World Characteristics

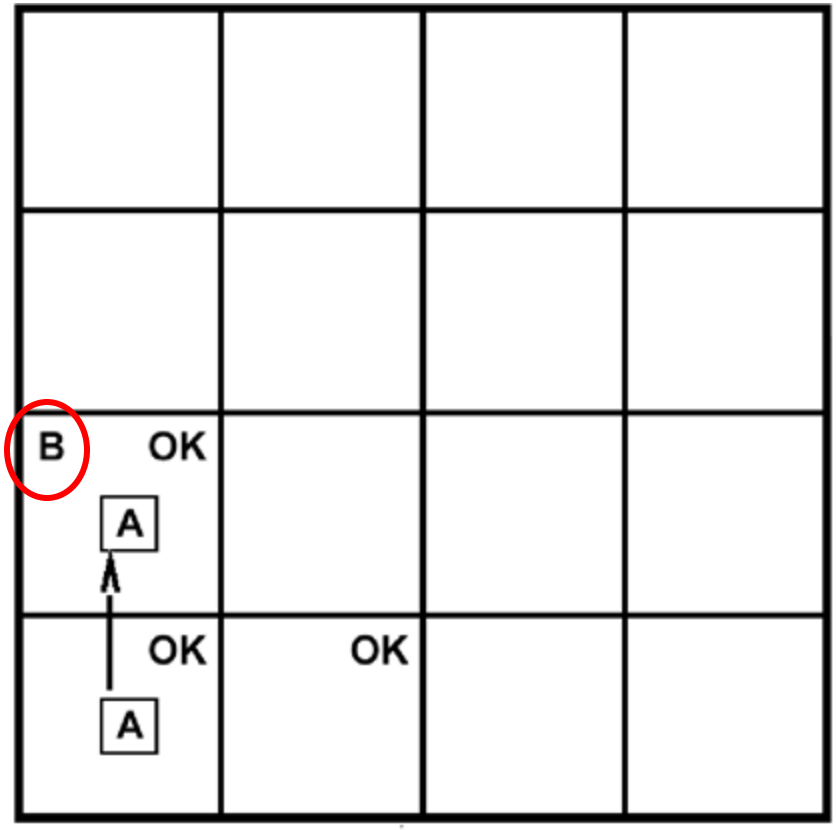
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  - → It must **infer** the map from local perception
- What's the belief state for this domain?
  - All possible configurations of the board (that haven't been ruled out by observations) → HUGE

# Exploring a Wumpus World



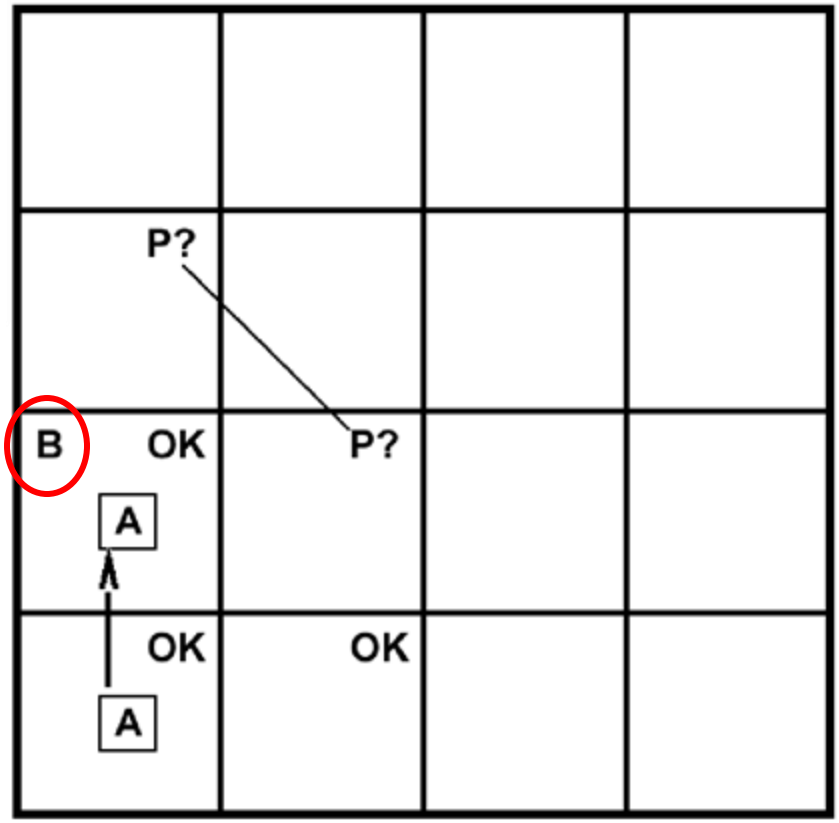

A= Agent
B= Breeze
S= Smell
P= Pit
W= Wumpus
OK = Safe
G = Glitter

# Exploring a Wumpus World



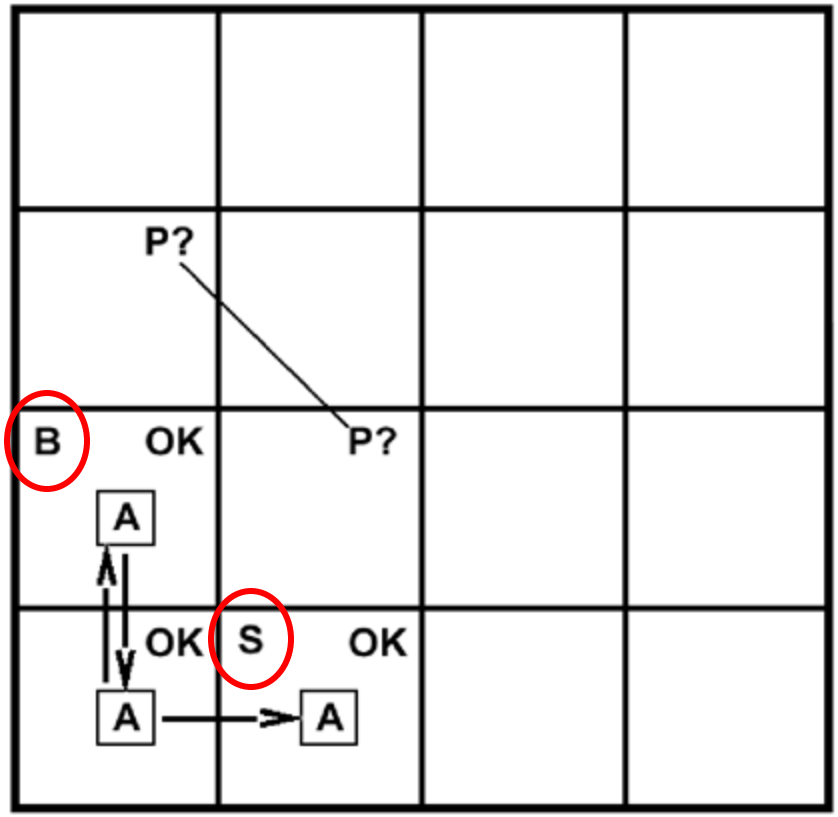

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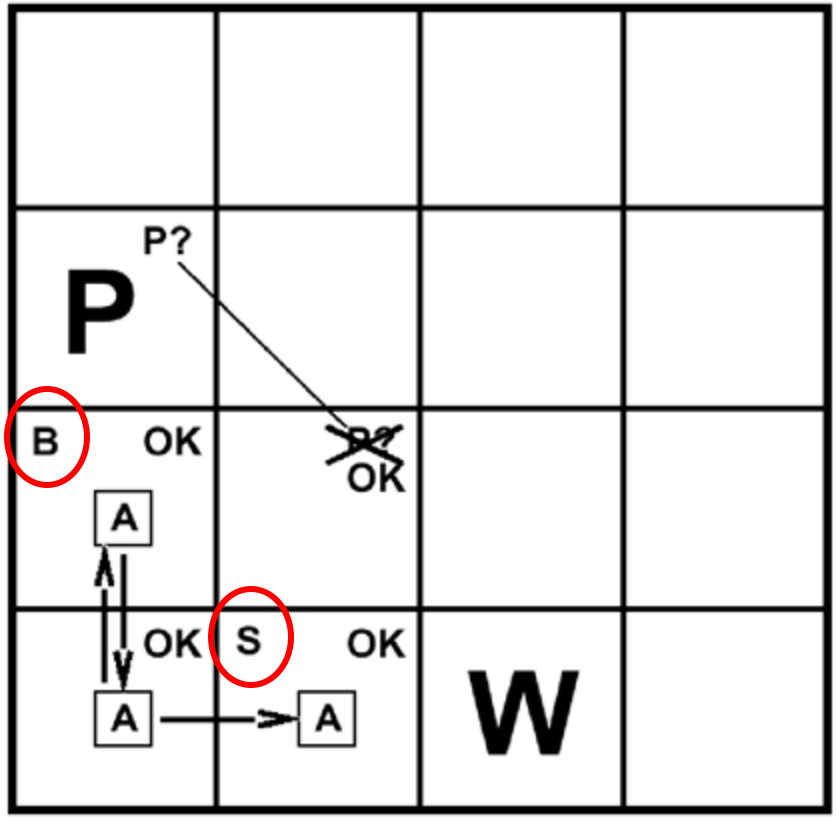

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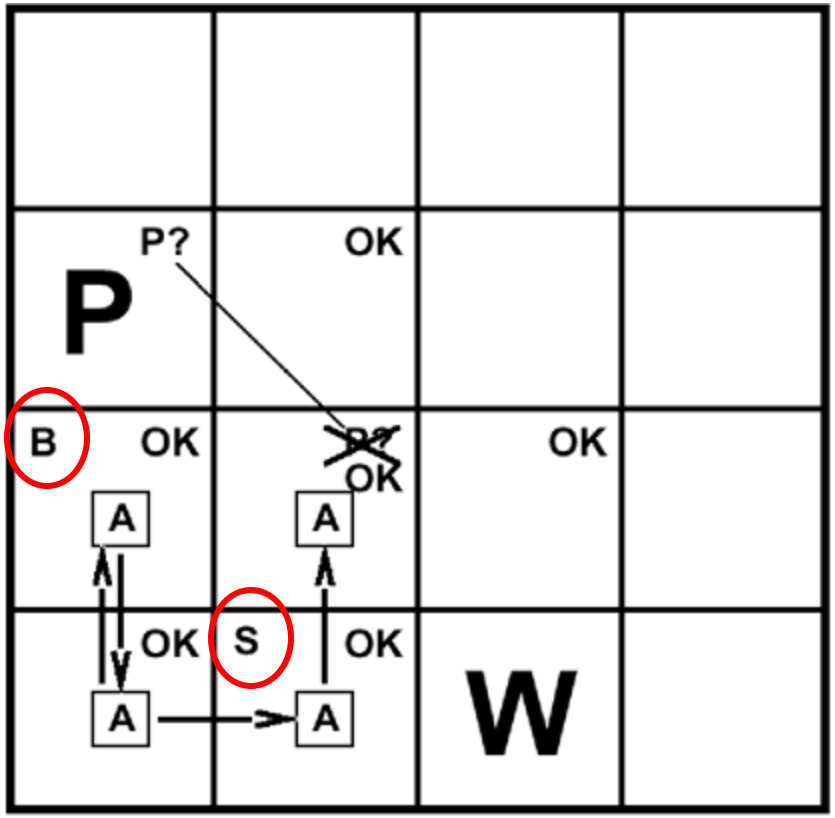

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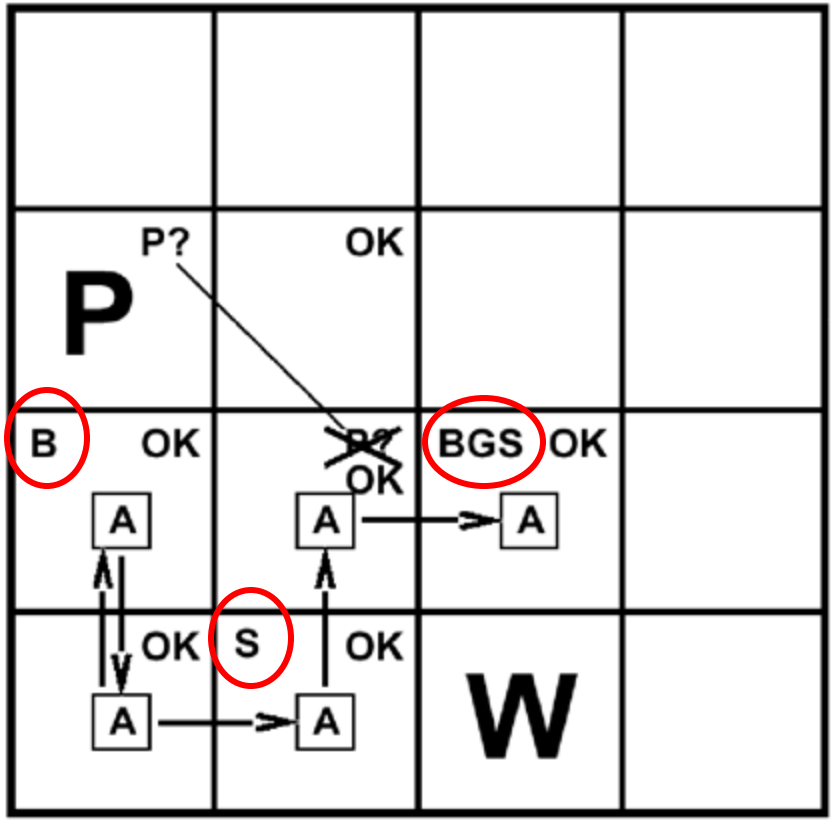

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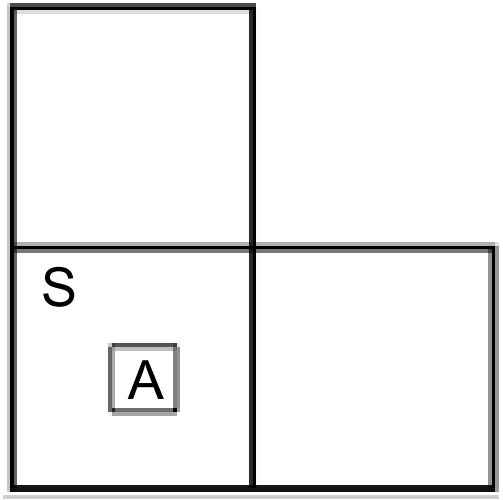
# Exploring a Wumpus World




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# Using actions to gain knowledge

- Suppose we enter another Wumpus cave
- Perceive *Smell* in (1, 1) ⇒ cannot move
- Use a strategy of coercion:
  - Shoot straight ahead
  - Wumpus was there ⇒ dead ⇒ safe
  - Wumpus wasn't there ⇒ safe




Logic is a type of **knowledge representation** that supports this reasoning!

# Logic

- Logics are **formal languages** for representing information
- They enable conclusions to be drawn from what is known
- A logic is defined by the following:
  - The **syntax** defines which **sentences** are allowed in the language
  - The **semantics** define the "meaning" of sentences,

    i.e., the **truth** of a sentence in a given **world**

# Logic

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An analogy: The language of arithmetic

$x+2 \geq y$ is a sentence

$x2 + y >$ is not a sentence

$x+2 \geq y$ is true iff the number x+2 is no less than the number y

$x+2 \geq y$ is true in a **world** where x=7, y=1

$x+2 \geq y$ is false in a **world** where x=0, y=6

# Types of logic

- Logics are characterized by what they allow as "primitives"
  - Ontological commitment: what exists? (facts/objects/time/beliefs)
  - Epistemological commitment: what do we know? (states of knowledge)

| Language | Ontological commitment | Epistemological commitment |
|---|---|---|
| Propositional logic | facts | true/false/unknown |
| First-order logic | facts, objects, relations | true/false/unknown |
| Temporal logic | facts, objects, relations, times | true/false/unknown |
| Probability theory | facts | degree of belief [0, 1] |
| Fuzzy logic | degree of truth | degree of belief [0, 1] |

# Propositional logic

- **Propositions** are the basic elements
  - They are assertions about the state of the world/game/problem
  - They can be **true** or **false**
  - e.g. “Today is Thursday”, “Today is Wednesday”

- **Syntax** specifies the valid sentences in the logic
  - **Atomic sentences** consist of a single **proposition** symbol, like the sentence $P_1$ or $W_{1,3}$
  - **Complex sentences** are constructed from simpler sentences
  - We combine sentences using parentheses and **logical connectives**

# Propositional logic: Syntax

There are five standard logical connectives:

| negation: ¬ | conjunction: ∧ | disjunction: ∨ | implication: ⇒ | biconditional: ⇔ |
|---|---|---|---|---|
| *not* | *and* | *or* | *implies* | *iff* |

**Syntax** defines the allowable sentences:

- Atomic sentences are sentences
- If $S$ is a sentence, $\neg S$ is a sentence
- If $S_1$ and $S_2$ are sentences, $S_1 \wedge S_2$ is a sentence
- If $S_1$ and $S_2$ are sentences, $S_1 \vee S_2$ is a sentence
- If $S_1$ and $S_2$ are sentences, $S_1 \Rightarrow S_2$ is a sentence
- If $S_1$ and $S_2$ are sentences, $S_1 \Leftrightarrow S_2$ is a sentence

# Propositional logic: Syntax

Some examples

$W_{(3,1)}$ $W_{(3,2)}$ $P_{(2,2)}$ $B_{(2,1)}$

$W_{(3,1)} \wedge W_{(3,2)}$

$P_{(2,\,2)} \Rightarrow B_{(2,\,1)}$

$(W_{(3,1)} \wedge W_{(3,2)}) \vee (P_{(2,2)} \Rightarrow B_{(2,1)})$

# Propositional logic: Semantics

- The **semantics** defines the rules for determining the truth of a sentence

- Truth is always determined ***with respect to*** a **model**
- A **model** is an assignment of truth values, one for every proposition symbol. e.g.,

$$m = \{P_1 = true, P_2 = false, P_3 = true\}$$

- A sentence may be **true** in one model and **false** in another

# Propositional logic: Semantics

- The **semantics** defines the rules for determining the truth of a sentence, with respect to a particular **model**

- Rules for evaluating truth with respect to model $m$:

| | | | | | |
|---|---|---|---|---|---|
| $\neg S$ | is true iff | $S$ is false | | | in $m$ |
| $S_1 \wedge S_2$ | is true iff | $S_1$ is true | and | $S_2$ is true | in $m$ |
| $S_1 \vee S_2$ | is true iff | $S_1$ is true | or | $S_2$ is true | in $m$ |
| $S_1 \Rightarrow S_2$ | is true iff | $S_1$ is false | or | $S_2$ is true | in $m$ |
| $S_1 \Leftrightarrow S_2$ | is true iff | $S_1 \Rightarrow S_2$ is true | and | $S_2 \Rightarrow S_1$ is true | in $m$ |

$S_1 \Rightarrow S_2$ can also be written as $\neg S_1 \vee S_2$

$S_1 \Leftrightarrow S_2$ can also be written as $(\neg S_1 \vee S_2) \wedge (S_1 \vee \neg S_2)$

# Interpretations: True or False

Some terminology:

- A sentence is **valid** if it is true in *all* models
- A sentence is **satisfiable** if it is true in *at least one* model
- A sentence is **unsatisfiable** if it is false in *all* models

- The truth of a sentence may depend on the model

e.g., Sentence = "*Today is Wednesday.*"

Is this **valid**? Is it **satisfiable**?

# Exercise

- For each of the following logical statements, say whether it is **valid**, **satisfiable** or **unsatisfiable**:

$A \vee \neg B$

$(A \wedge B) \vee (\neg A \vee \neg B)$

$\neg A \Rightarrow A$

$B \wedge (B \Rightarrow \neg B)$

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True ∨ ¬True

True ∨ ¬False

False ∨ ¬True

False ∨ ¬False

# Exercise

- For each of the following logical statements, say whether it is **valid**, **satisfiable** or **unsatisfiable**:

$A \vee \neg B$ satisfiable

True ∨ ¬True
True ∨ ¬False
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satisfiable

$(A \wedge B) \vee (\neg A \vee \neg B)$

(True ∧ True) ∨ (¬True ∨ ¬True)
(True ∧ False) ∨ (¬True ∨ ¬False)
(False ∧ True) ∨ (¬False ∨ ¬True)
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- For each of the following logical statements, say whether it is **valid**, **satisfiable** or **unsatisfiable**:

$A \vee \neg B$ — satisfiable

$(A \wedge B) \vee (\neg A \vee \neg B)$ — valid

(True ∧ True) ∨ (¬True ∨ ¬True)
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| $\neg A \Rightarrow A$ | satisfiable | ¬True ⇒ True<br>¬False ⇒ False |
| $B \wedge (B \Rightarrow \neg B)$ | | |

# Exercise

- For each of the following logical statements, say whether it is **valid**, **satisfiable** or **unsatisfiable**:

| | |
|---|---|
| $A \vee \neg B$ | satisfiable |
| $(A \wedge B) \vee (\neg A \vee \neg B)$ | valid |
| $\neg A \Rightarrow A$ | satisfiable |
| $B \wedge (B \Rightarrow \neg B)$ | True ∧ (True ⇒ ¬True)<br>False ∧ (False ⇒ ¬False) |

# Exercise

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| $(A \wedge B) \vee (\neg A \vee \neg B)$ | valid | |
| $\neg A \Rightarrow A$ | satisfiable | |
| $B \wedge (B \Rightarrow \neg B)$ | unsatisfiable | True ∧ (True ⇒ ¬True)<br>False ∧ (False ⇒ ¬False) |

# Using logic

- We want to derive **sound conclusions** from what we already know

- Thus, we want a process for generating new sentences that are **guaranteed to be true**
  - (in any world in which the premises are true)

- The first step is the concept of **entailment:** $\alpha \models \beta$
  - We say $\alpha$ entails $\beta$
  - Or, $\beta$ follows logically from $\alpha$

# Entailment

- Sentence $\alpha$ **entails** sentence $\beta$ if and only if $\beta$ is true in all worlds where $\alpha$ is true:

$$\alpha \models \beta \text{ if and only if } M(\alpha) \subseteq M(\beta)$$

The set of all models in which $\alpha$ is true.
"The set of all models that satisfy $\alpha$."

- $\alpha$ is a stronger assertion than $\beta$, since $\alpha$ rules out more possible worlds

- Another arithmetic illustration:

sentence $x = 0$ **entails** sentence $xy = 0$

# Entailment in Wumpus World

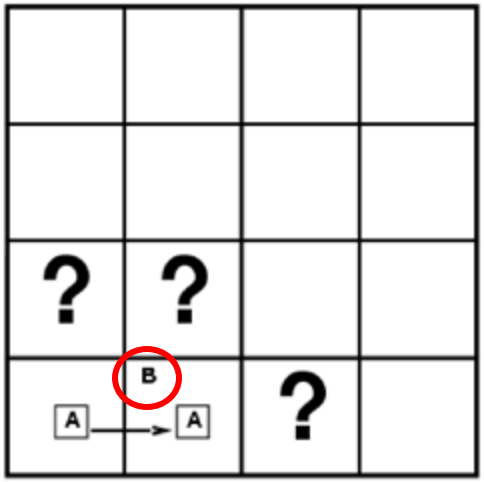
Let's consider this situation:

- The agent detects *Nothing* in [1,1], moves right, and detects *Breeze* in [2,1]

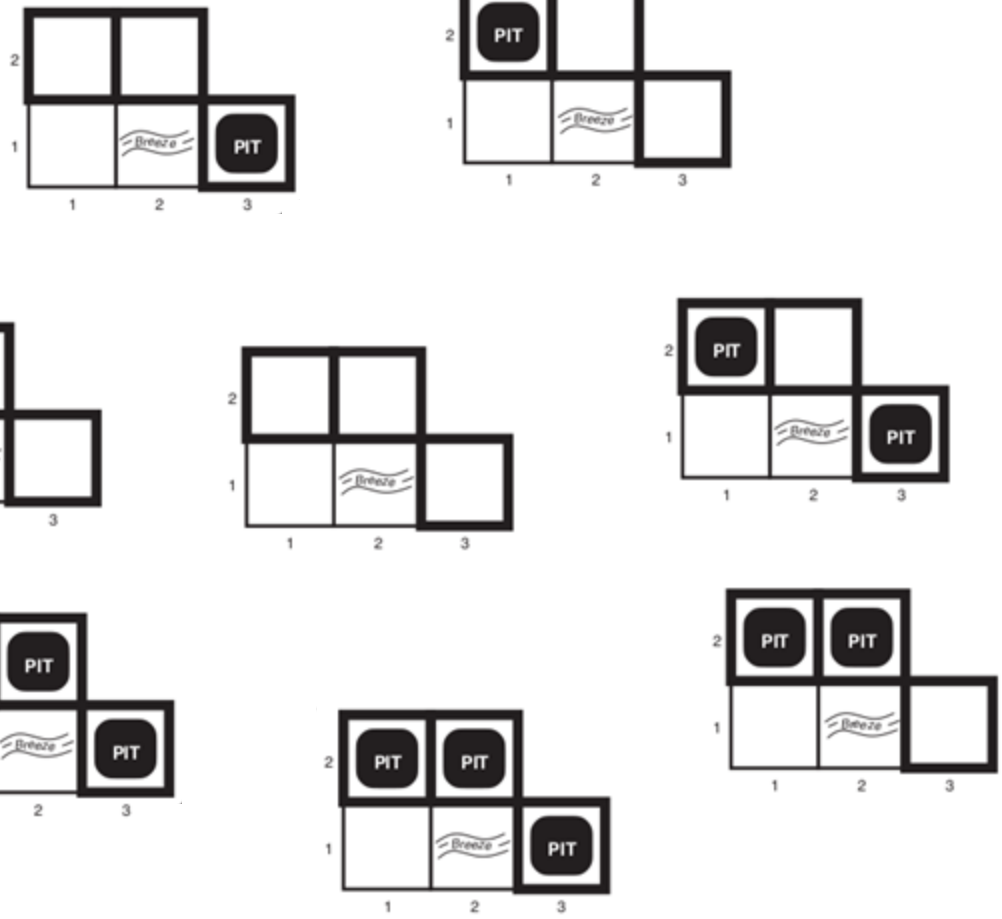
- The agent wants to **infer** the presence of pits in squares [1,2], [2,2], and [3,1]

- There are 3 Boolean (true-false) choices:
  - $P_{[1,2]} = true$ $\quad P_{[1,2]} = false$
  - $P_{[2,2]} = true$ $\quad P_{[2,2]} = false$
  - $P_{[3,1]} = true$ $\quad P_{[3,1]} = false$

  $\Rightarrow$ 8 possible models

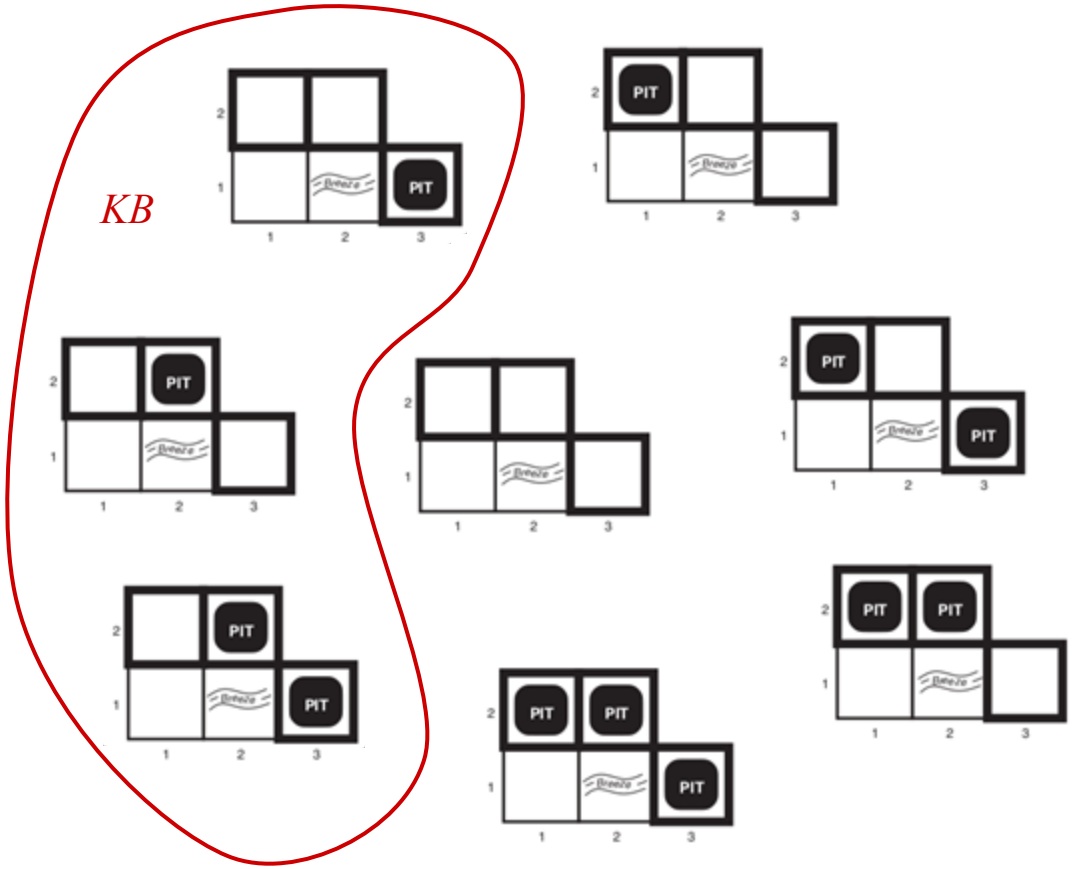


# Models in Wumpus World



# KB in Wumpus World

- KB = wumpus-world rules + observations

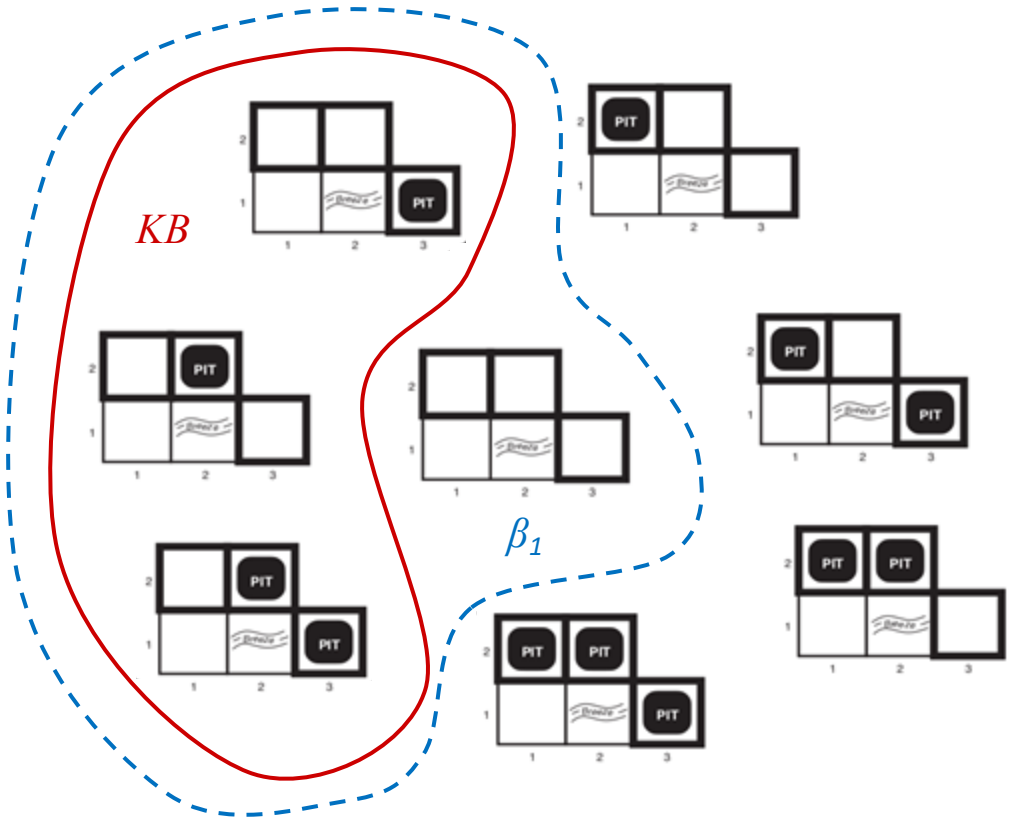


# Model checking in Wumpus World

- KB = wumpus-world rules + observations
- Consider $\beta_1$ = "square [1,2] is safe"
- Is it the case that KB $\models \beta_1$ ?
- Let's use **model checking** => Enumerate models and check

# Model checking in Wumpus World

- KB = wumpus-world rules + observations
- Consider $\beta_1$ = "square [1,2] is safe"




By model checking,
KB ⊨ $\beta_1$ 😊

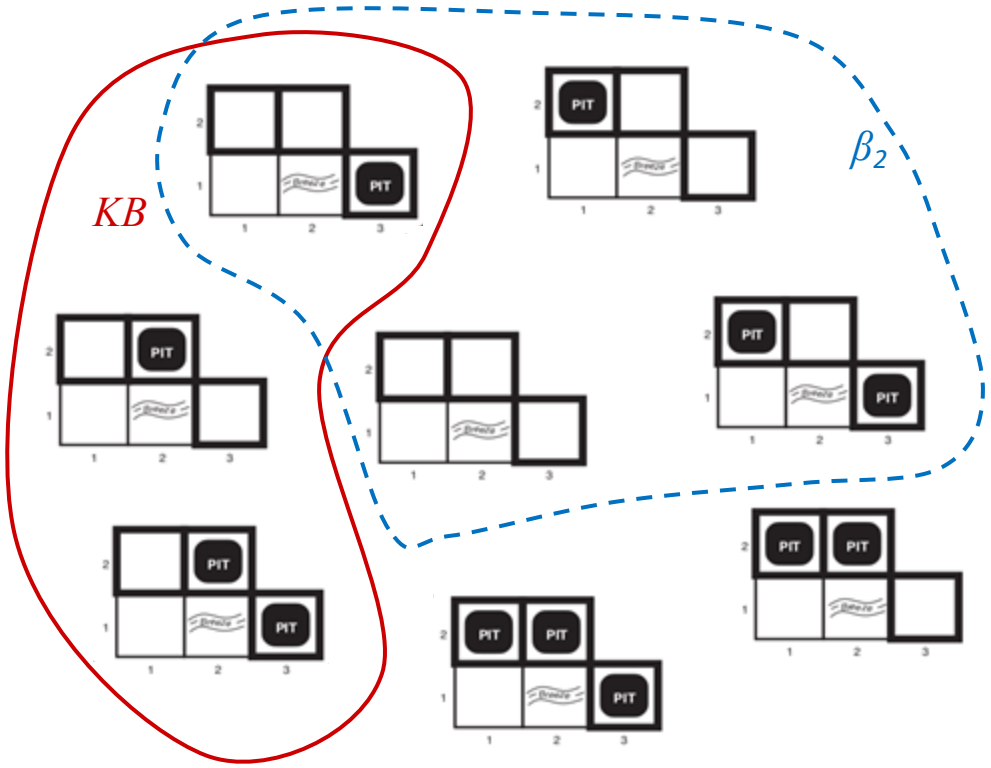
Recall:
*KB entails $\beta$*
*if and only if*
*$\beta$ is true in all worlds*
*where KB is true.*

# Model checking in Wumpus World

- KB = wumpus-world rules + observations
- Consider $\beta_2$ = "square [2,2] is safe"
- Is it the case that KB $\models \beta_2$ ?

# Model checking in Wumpus World

- KB = wumpus-world rules + observations
- Consider $\beta_2$ = "square [2,2] is safe"




By model checking,
KB $\not\models \beta_2$ 😞

Recall:
*KB <u>entails</u> $\beta$*
*if and only if*
*$\beta$ is true in all worlds*
*where KB is true.*

# The Knowledge-based Agent

- Recall our agent, with its **knowledge base** and **inference procedure**

- Now we have an idea for what the KB actually is:
  - a set of sentences in propositional logic!

- We can also see the utility of entailment

  If $KB \models \beta$ then:
  - in every model in which $KB$ is true, $\beta$ is also true
  - we can derive the conclusion $\beta$ from what we already know (the $KB$)



The process of logical inference!

# Inference procedure

- $KB \vdash_i \beta$

  means sentence $\beta$ can be derived from $KB$ by inference procedure $i$

- Desired qualities of inference procedure $i$:

  **Soundness**: $i$ is sound if, whenever $KB \vdash_i \beta$ is true, it is also true that $KB \models \beta$.

  In other words, *we only infer entailed sentences.*

  **Completeness**: $i$ is complete if, whenever $KB \models \beta$ is true, it is also true that $KB \vdash_i \beta$.

  In other words, *we can generate all the entailed sentences*.

# Two inference methods

1. Model checking:
   - Truth table enumeration

2. Application of inference rules

# Propositional inference

Truth table method:

Let $\beta = A \vee B$ and $KB = (A \vee C) \wedge (B \vee \neg C)$

Is it the case that $KB \models \beta$ ?

Check all possible models: $\beta$ must be true whenever *KB* is true.

| *A* | *B* | *C* | *A* ∨ *C* | *B* ∨ ¬*C* | *KB* | $\beta = A \vee B$ |
|---|---|---|---|---|---|---|
| *False* | *False* | *False* | | | | |
| *False* | *False* | *True* | | | | |
| *False* | *True* | *False* | | | | |
| *False* | *True* | *True* | | | | |
| *True* | *False* | *False* | | | | |
| *True* | *False* | *True* | | | | |
| *True* | *True* | *False* | | | | |
| *True* | *True* | *True* | | | | |

# Propositional inference

Truth table method:

Let $\beta = A \vee B$ and $KB = (A \vee C) \wedge (B \vee \neg C)$

Is it the case that $KB \models \beta$ ?

Check all possible models: $\beta$ must be true whenever *KB* is true.

| *A* | *B* | *C* | *A ∨ C* | *B ∨ ¬C* | *KB* | $\beta = A \vee B$ |
|---|---|---|---|---|---|---|
| *False* | *False* | *False* | False | True | False | False |
| *False* | *False* | *True* | True | False | False | False |
| *False* | *True* | *False* | False | True | False | True |
| *False* | *True* | *True* | True | True | True | True |
| *True* | *False* | *False* | True | True | True | True |
| *True* | *False* | *True* | True | False | False | True |
| *True* | *True* | *False* | True | True | True | True |
| *True* | *True* | *True* | True | True | True | True |

# Propositional inference

Truth table method:

Let $\beta = A \vee B$ and $KB = (A \vee C) \wedge (B \vee \neg C)$

Is it the case that $KB \models \beta$ ? **YES!**

Check all possible models: $\beta$ must be true whenever *KB* is true.

| *A* | *B* | *C* | *A* ∨ *C* | *B* ∨ ¬*C* | *KB* | $\beta = A \vee B$ |
|---|---|---|---|---|---|---|
| *False* | *False* | *False* | False | True | False | False |
| *False* | *False* | *True* | True | False | False | False |
| *False* | *True* | *False* | False | True | False | True |
| *False* | *True* | *True* | True | True | True | True |
| *True* | *False* | *False* | True | True | True | True |
| *True* | *False* | *True* | True | False | False | True |
| *True* | *True* | *False* | True | True | True | True |
| *True* | *True* | *True* | True | True | True | True |

# Properties of the truth table method

- Is it sound? Yes.
- Is it complete? Yes. It checks truth in all possible models.
- Time / space complexity?
  - Highly inefficient! $2^n$ models for $n$ literals!
- A **literal** is an atomic sentence (positive literal) or a negated atomic sentence (negative literal)

# Two kinds of inference methods

1. Model checking:
   - Truth table enumeration (sound and complete, but expensive)

1. Application of inference rules (*theorem proving*):
   - A **proof** is a sequence of inference rule applications that generates the desired sentence
   - We can use inference rules as operators in a standard search algorithm to **automate** theorem proving

# Inference rules for propositional logic

- **Resolution:**

$$\frac{(\alpha \vee \beta), \quad (\neg\beta \vee \gamma)}{(\alpha \vee \gamma)}$$

- **Modus Ponens:**

$$\frac{\alpha_1, \alpha_2, \ldots \alpha_n, \quad (\alpha_1 \wedge \alpha_2 \wedge \ldots \wedge \alpha_n \Rightarrow \beta)}{\beta}$$

- **Contraposition:**

$$\frac{\alpha \Rightarrow \beta}{\neg\beta \Rightarrow \neg\alpha}$$

# Other useful rules

- **And-elimination:**

$$\frac{\alpha_1 \wedge \alpha_2 \wedge \ldots \wedge \alpha_n}{\alpha_1,\ \alpha_2,\ \ldots,\ \alpha_n}$$

- **Implication elimination:**

$$\frac{\alpha \Rightarrow \beta}{\neg\alpha \vee \beta}$$

- **De Morgan's law:**

$$\frac{\neg(\alpha \vee \beta)}{(\neg\alpha) \wedge (\neg\beta)} \qquad \frac{\neg(\alpha \wedge \beta)}{(\neg\alpha) \vee (\neg\beta)}$$

# Searching with inference rules

- Inference rules can be used with **forward search** or **backward search**
- This brings the algorithms we've seen previously, like iterative deepening, into play for **theorem proving**
- Thus, **logic enables us to automate reasoning**

# Forward chaining

- When a new sentence *p* is added to the KB
  - Look for all sentences that share literals with *p*
  - Perform inference
  - Add new sentences to the KB and continue until *p* is added or more inferences can be made

- Two important properties:
  - Forward chaining is data-driven:

    It infers properties and categories for new percepts
  - Forward chaining is an eager method: new facts are inferred as soon as possible

# Backward chaining

- When a query *q* is asked of the KB
  - If *q* is in the KB already, return true
  - Else, use inference for *q* with other sentences in the KB to find implications in the KB whose conclusion is *q*
    - If all the premises of one of those implications is true, then *q* is true

- Two important properties:
  - Backward chaining is goal-driven:

    It centers the reasoning around the query being asked
  - It is a lazy reasoning method: new facts are only inferred as needed, and only to the extent that they help answer the question

# Forward chaining example

Let's say we have a knowledge base with these sentences:

- HaveAILecture ⇒ (TodayIsMonday ∨ TodayIsWednesday)
- ¬TodayIsMonday
- HaveAILecture ∨ HaveNoClass
- HaveNoClass ⇒ Sad
- ¬Sad

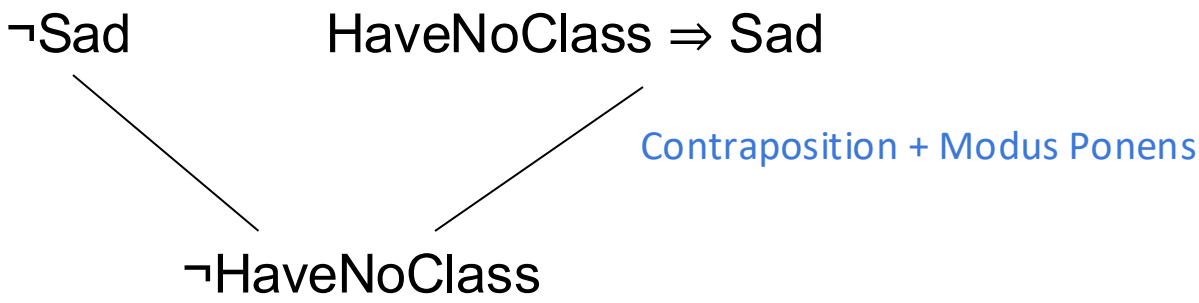
**Can you infer what day it is?**

# Forward chaining example

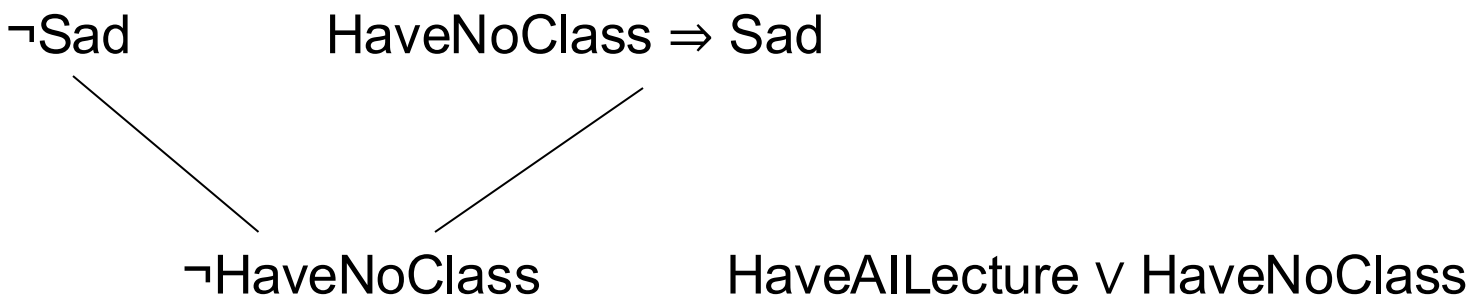
¬Sad

HaveNoClass ⇒ Sad

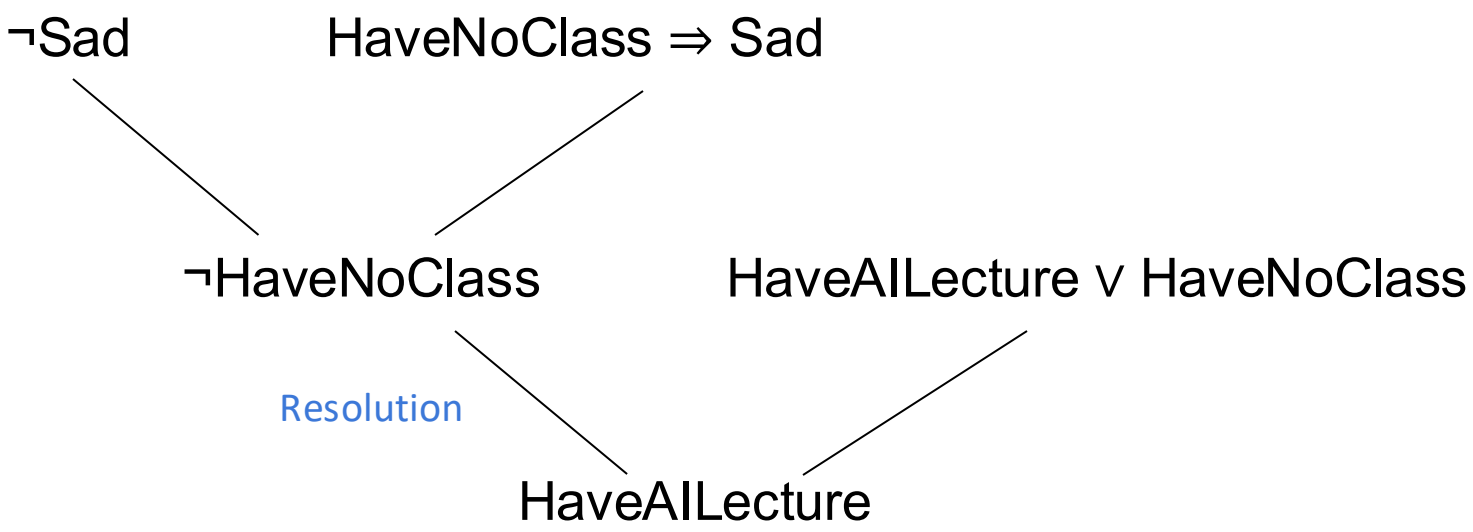
# Forward chaining example



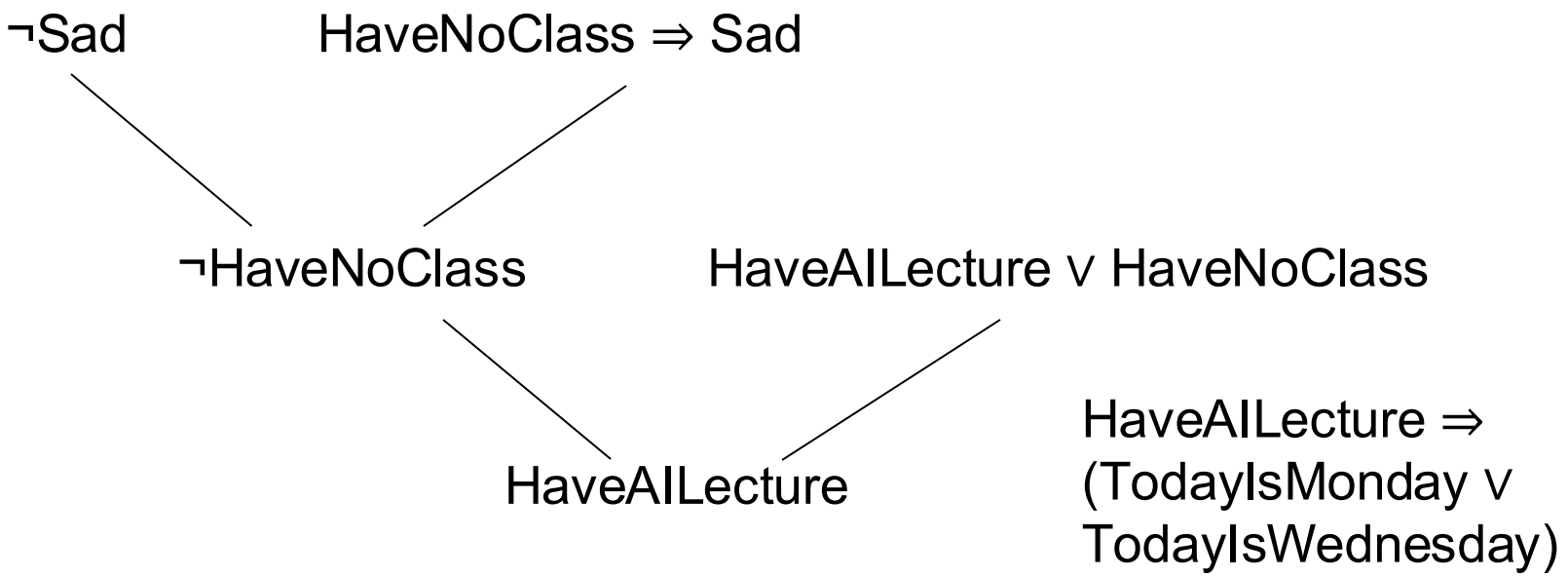
# Forward chaining example



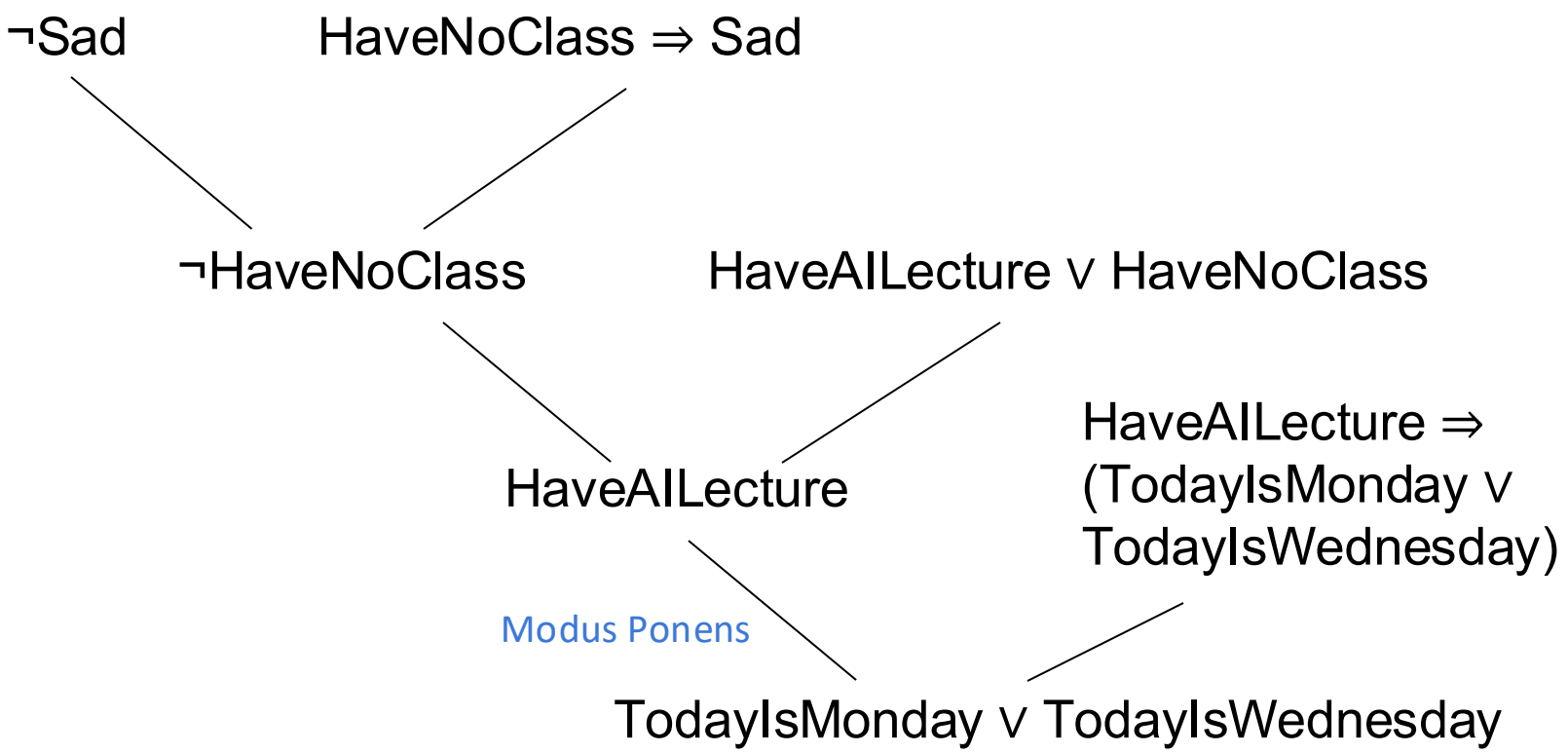
# Forward chaining example



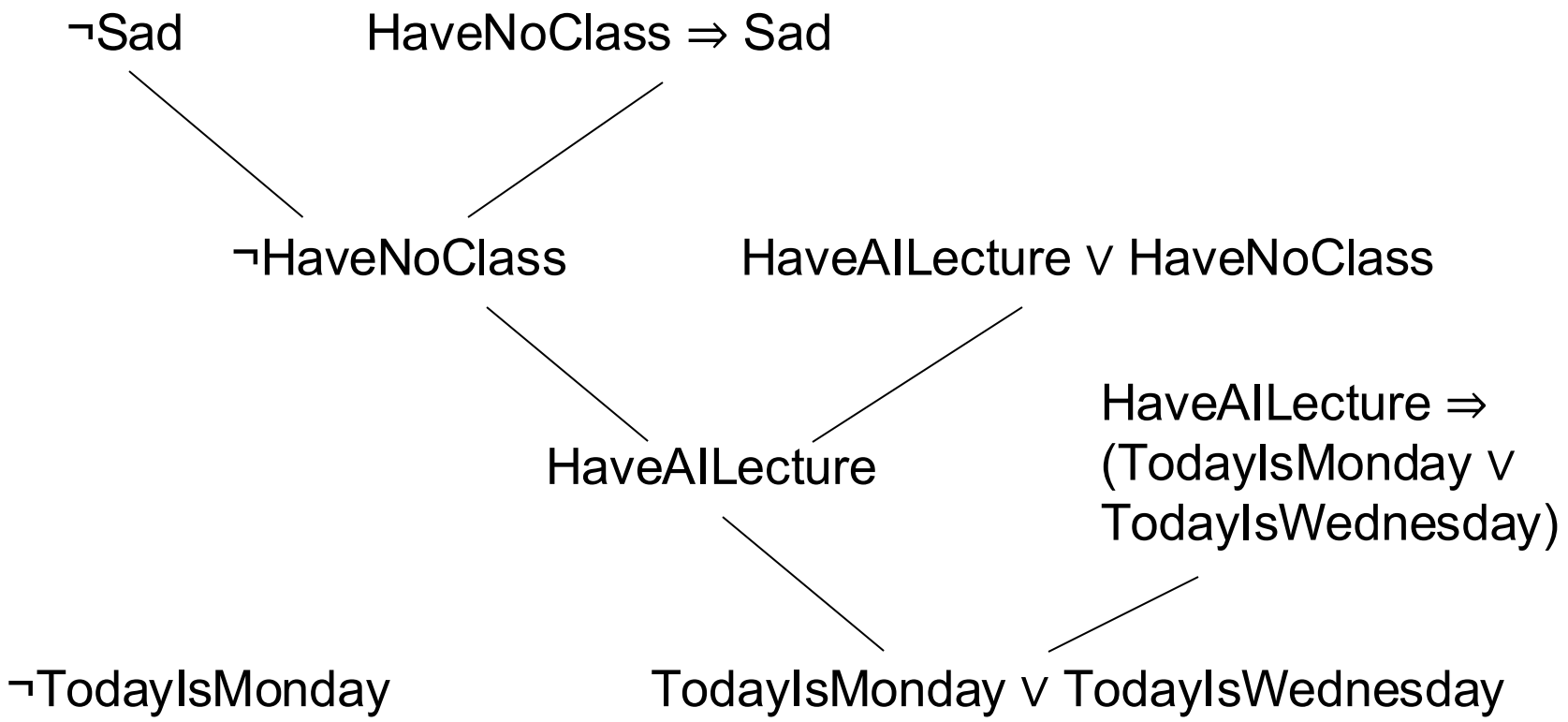
# Forward chaining example



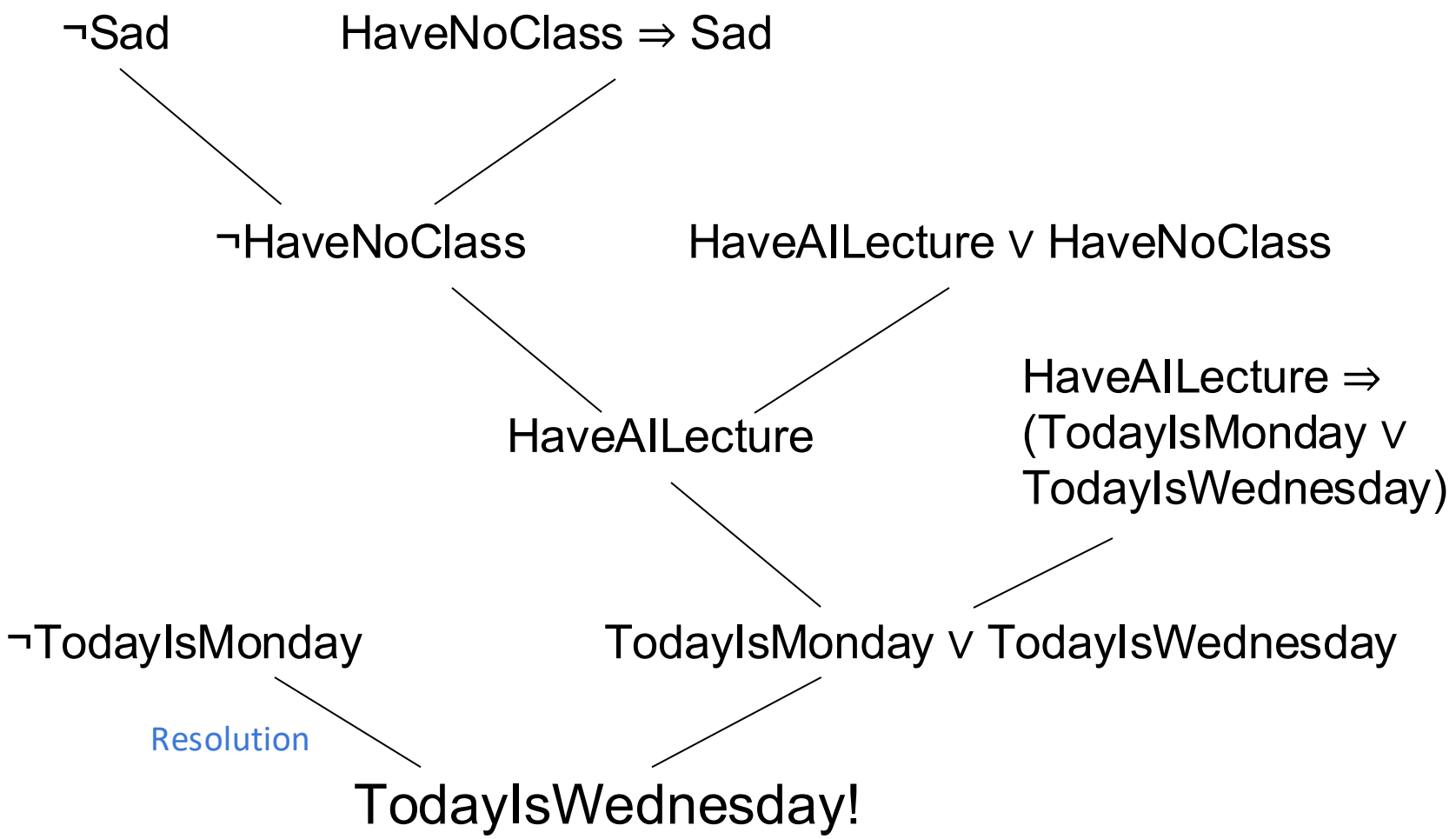
# Forward chaining example



# Forward chaining example



# Forward chaining example



# Summary

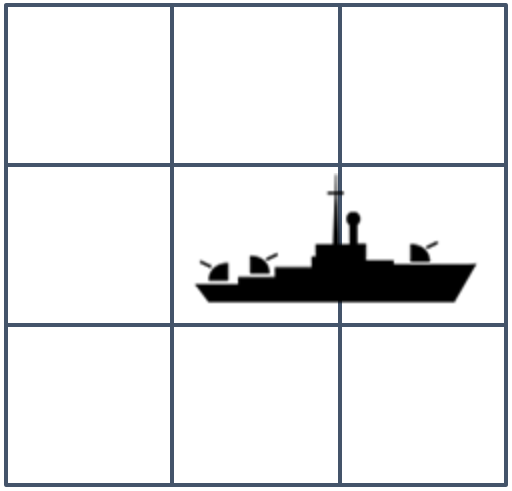
- A knowledge-based agent
  - Knowledge base
  - Inference engine
- Wumpus World

- Propositional Logic
  - Sentences and Models
  - Entailment

- Inference methods
  - Model checking
  - Theorem proving
    - Inference rules
    - Forward chaining and backward chaining

# The Goal Today





# The Goal Today

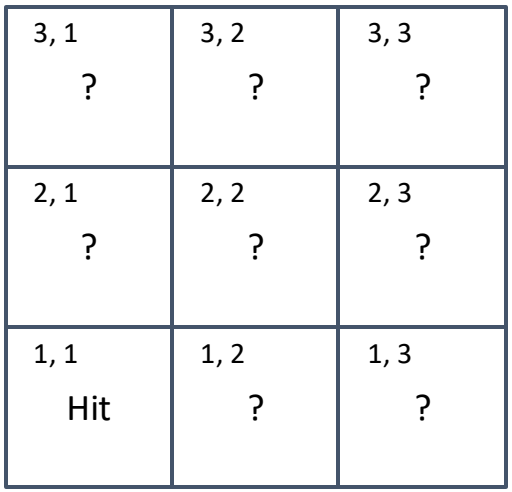
| | | |
|---|---|---|
| 3, 1 ? | 3, 2 ? | 3, 3 ? |
| 2, 1 ? | 2, 2 ? | 2, 3 ? |
| 1, 1 ? | 1, 2 ? | 1, 3 ? |

# The Goal Today

| | | |
|---|---|---|
| 3, 1 ? | 3, 2 ? | 3, 3 ? |
| 2, 1 ? | 2, 2 ? | 2, 3 ? |
| 1, 1 ? | 1, 2 ? | 1, 3 ? |

$$B_{1,1} \Rightarrow (B_{1,2} \vee B_{2,1})$$
$$B_{1,2} \Rightarrow (B_{1,1} \vee B_{1,3} \vee B_{2,2})$$
$$B_{1,3} \Rightarrow (B_{1,2} \vee B_{2,3})$$
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$$B_{3,3} \Rightarrow (B_{3,2} \vee B_{2,3})$$

# The Goal Today




$B_{1,1} \Rightarrow (B_{1,2} \vee B_{2,1})$
$B_{1,2} \Rightarrow (B_{1,1} \vee B_{1,3} \vee B_{2,2})$
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$B_{3,3} \Rightarrow (B_{3,2} \vee B_{2,3})$

$B_{1,1}$

# The Goal Today

| | | |
|---|---|---|
| 3, 1 ? | 3, 2 ? | 3, 3 ? |
| 2, 1 ? | 2, 2 ? | 2, 3 ? |
| 1, 1 Hit | 1, 2 ? | 1, 3 ? |

$B_{1,1} \Rightarrow (B_{1,2} \vee B_{2,1})$
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$B_{1,1}$
$B_{1,2} \vee B_{2,1}$

# The Goal Today

| | | |
|---|---|---|
| 3, 1 ? | 3, 2 ? | 3, 3 ? |
| 2, 1 Empty | 2, 2 ? | 2, 3 ? |
| 1, 1 Hit | 1, 2 ? | 1, 3 ? |

$B_{1,1} \Rightarrow (B_{1,2} \vee B_{2,1})$
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$B_{1,1}$
$B_{1,2} \vee B_{2,1}$

$\neg B_{2,1}$

# The Goal Today

| | | |
|---|---|---|
| 3, 1 ? | 3, 2 ? | 3, 3 ? |
| 2, 1 Empty | 2, 2 ? | 2, 3 ? |
| 1, 1 Hit | 1, 2 ? | 1, 3 ? |

$B_{1,1} \Rightarrow (B_{1,2} \vee B_{2,1})$
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$B_{3,3} \Rightarrow (B_{3,2} \vee B_{2,3})$

$B_{1,1}$
$B_{1,2} \vee B_{2,1}$

$\neg B_{2,1}$
$B_{1,2}$

# The Goal Today

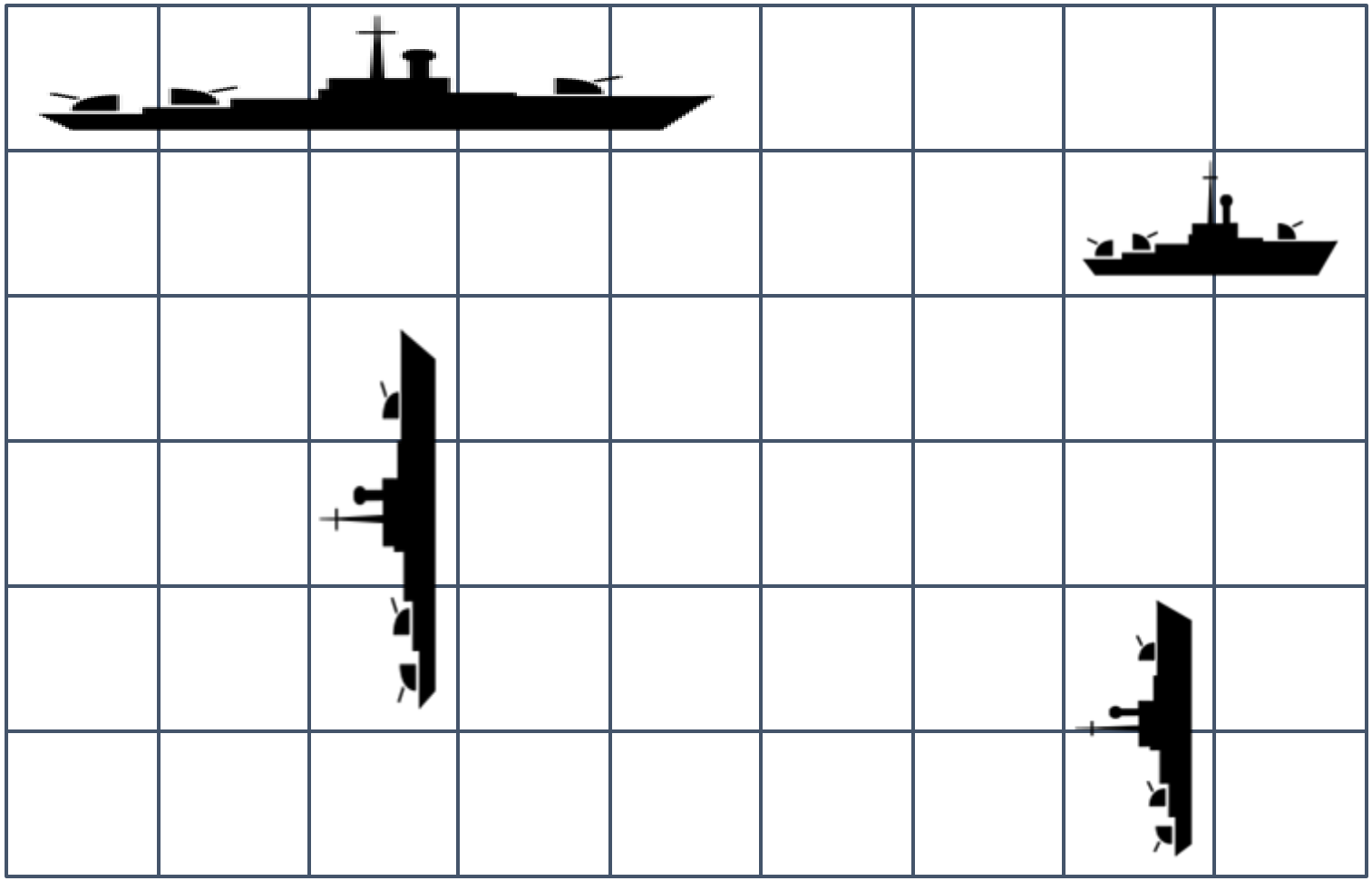
| | | |
|---|---|---|
| 3, 1<br>? | 3, 2<br>? | 3, 3<br>? |
| 2, 1<br>Empty | 2, 2<br>? | 2, 3<br>? |
| 1, 1<br>Hit | 1, 2<br>Hit | 1, 3<br>? |

$B_{1,1} \Rightarrow (B_{1,2} \vee B_{2,1})$
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$B_{3,3} \Rightarrow (B_{3,2} \vee B_{2,3})$

$B_{1,1}$
$B_{1,2} \vee B_{2,1}$

$\neg B_{2,1}$
$B_{1,2}$

# The Goal Today



# Next Class

- First-Order Logic